

TABLE OF CONTENTS

TABLE OF CONTENTS.....	1
1- OVERVIEW.....	2
2- TOOLS AND INSTRUMENTS REQUIRED.....	2
3- INITIAL CONDITIONS.....	3
4- HARDWARE PREPARATION.....	4
5- BODY CHECKS.....	9
6- HEAD CHECKS.....	15
7- AMPLIFIER SHUTDOWN VERIFICATION.....	23
8- DECLARATION FORM PREPARATION.....	25
9- SYSTEM RESTORATION.....	26
REVISION HISTORY.....	28

Description - This section applies to Signa systems installed with the 1.0T and 1.5T RF/PEN 2 cabinet.

1- OVERVIEW

Use this procedure to ensure that the power monitors are fully functional, including redundant trip capability. The tests described use a special pulse sequence data base (PSD) to provide excitation for the RF amplifier. The PSD allows the service engineer to selectively force peak power, pulse width, and duty cycle beyond allowable limits. Select setpoints for testing by modifying control variables that override the values normally calculated and downloaded by the PSD.

After checking the setpoints, the actual shutdown of the amplifier is verified three times: once for monitor A, once for monitor B, and once for the HV relay shutdown.

2- TOOLS AND INSTRUMENTS REQUIRED

See Table 2-1.

TABLE 2-1
TOOLS AND INSTRUMENTS REQUIRED (1.0T AND 1.5T)

ITEM	DESCRIPTION	PART NUMBER	QTY.
1.	RF Wattmeter - Bird Thruline Model 4391	46-255837P1	1
2.	10000-Watt interchangeable element Bird No. 10000B	46-255837p101	1
3.	500-Watt interchangeable element Bird No. 5000B	46-255837P105	1
4.	RF Test Cables Kit	46-255816G1	1
5.	Fifty-ohm, 200 watt, 30 dB attenuator - Bird Model 8322	46-255837P10	1
—OR—			
6.	RF Power Measurement Kit	46-317724G1 or G2	1
7.	100MHz Scope - Tektronix 468 or equivalent	46-183029P61 or P64	1

3- INITIAL CONDITIONS

Perform the prerequisite calibration procedures in the sequence listed in Table 3-1.

TABLE 3-1
POWER MONITOR CALIBRATION PROCEDURE

STEP	TAB	PROCEDURE	WHEN REQUIRED
1a or	RF	<i>Dummy Load & Cables Calibration</i>	If dummy load and cables have not been calibrated
1b	RF	<i>RF Power Measurement Calibration</i>	If oscilloscope has not been calibrated. (RF Power Measurement Kit)
2	RF	<i>APM Calibration</i>	Initial installation or if RF amp has been repaired
3	RF	<i>SAR Configuration File</i>	Initial installation or if site SAR limits change
4	RF	<i>Electrical Isocenter CAL (New DQA)</i>	Initial installation or if step 1 or 2 are redone



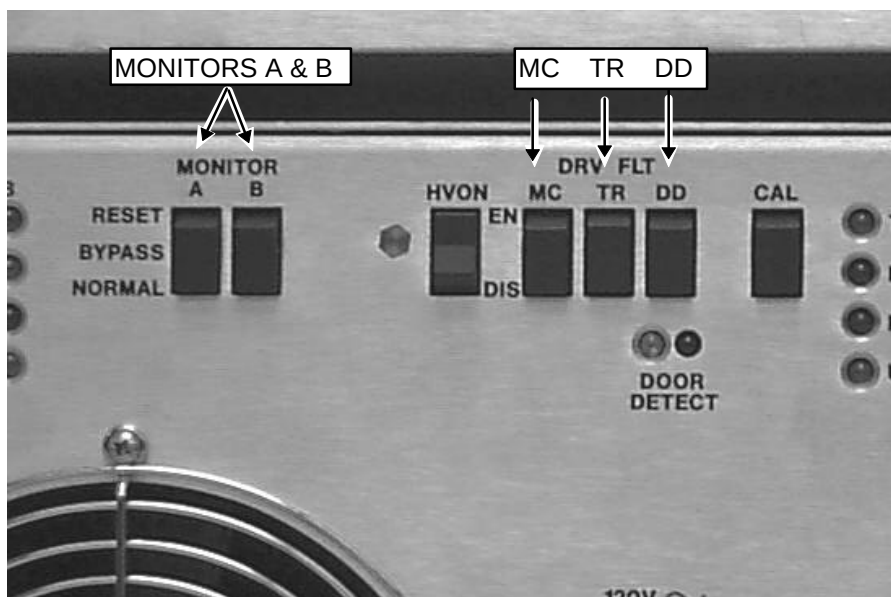
Possible equipment damage. Ensure that the RF amplifier has a dummy load connected to its output before initiating tests. If not, amplifier components could be permanently damaged.

Software Status

Signa software must be running, and the system must be capable of normal imaging.

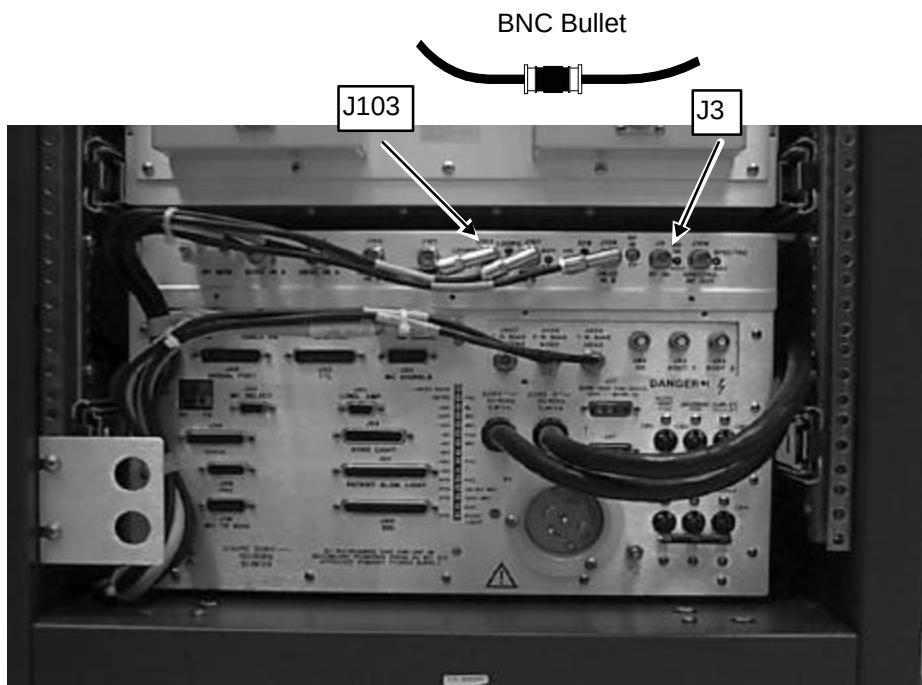
4- HARDWARE PREPARATION

1. Remove the front door from the cabinet.
2. On the front panel of the SSM, place the two monitor switches in the Normal position, and place the three DRV FLT switches in the disable faults position (DIS). See Illustration 4-1



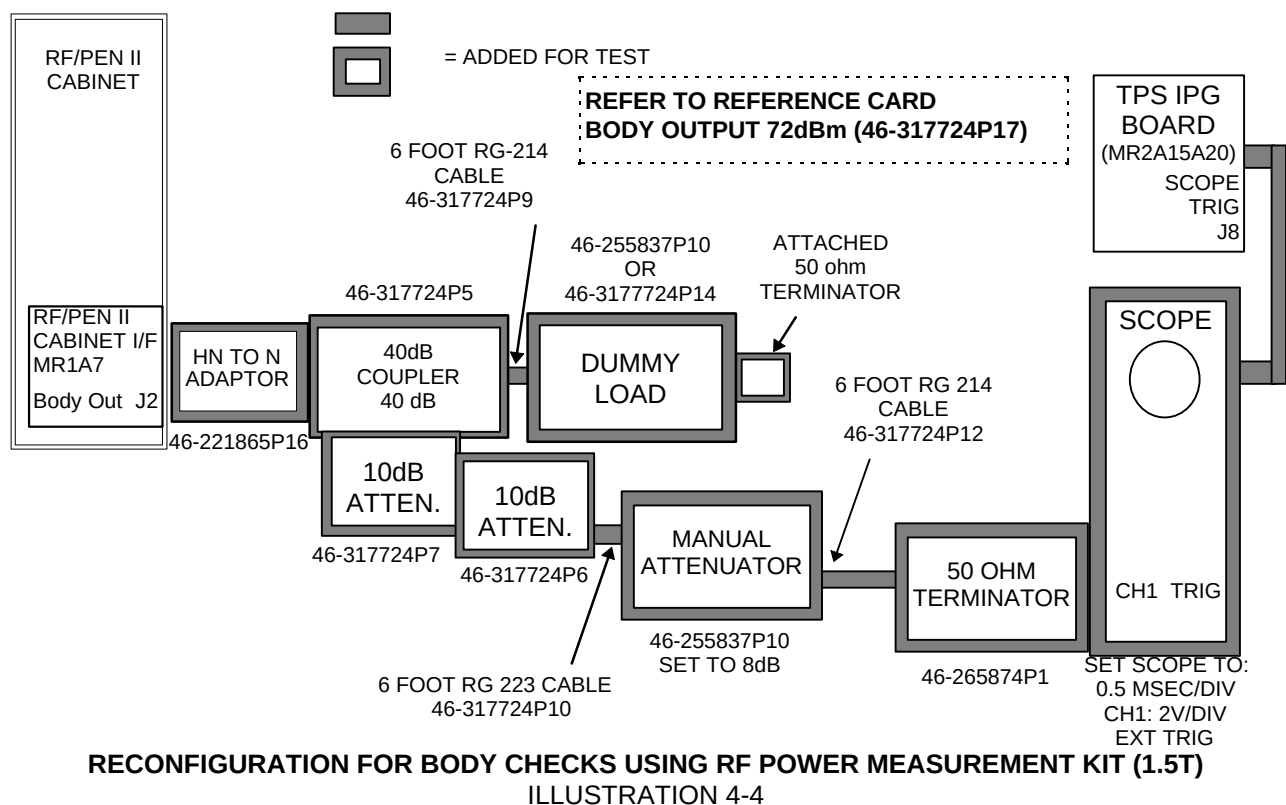
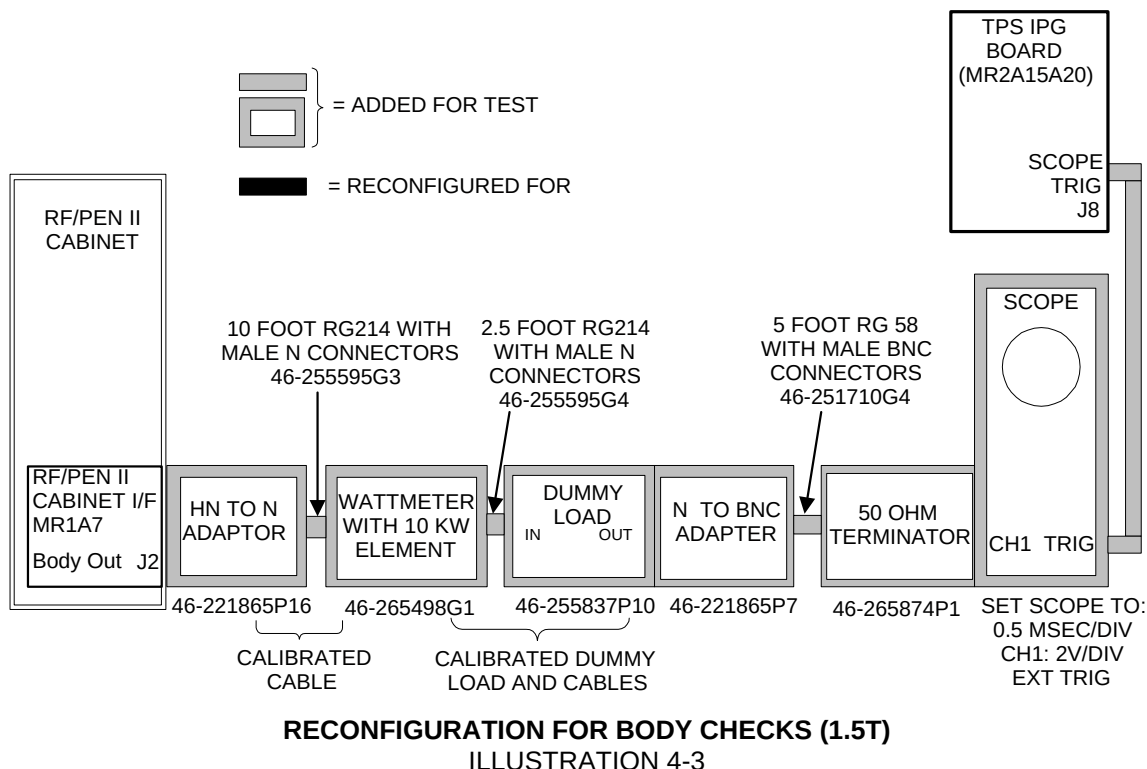
FRONT PANEL SWITCHES DISABLED
ILLUSTRATION 4-1

3. Bypass EFB with a BNC Bullet inline, between RF IN (J3) and EFB RF OUT (J103) on the rear panel of the system support module (SSM). See Illustration 4-2.

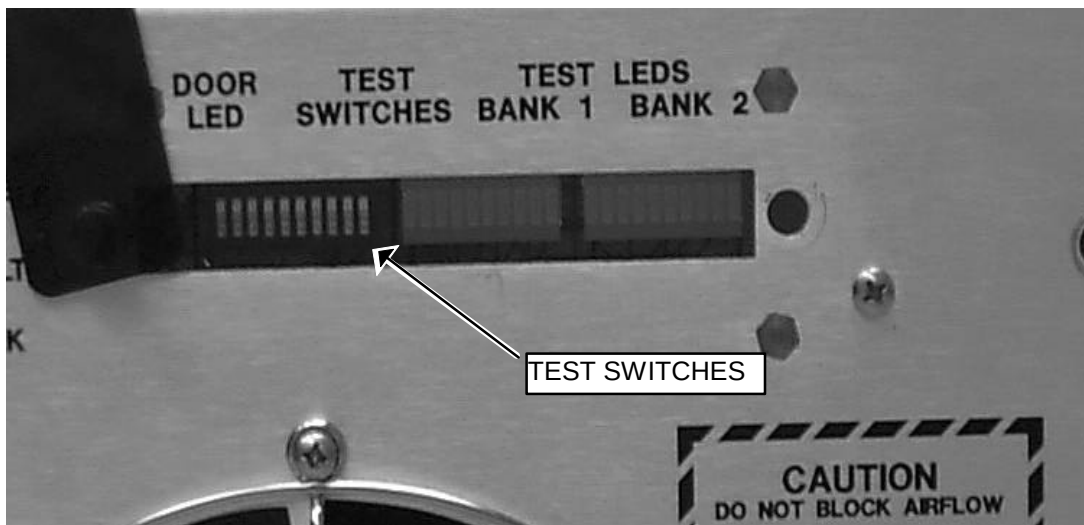


BYPASSING EFB
ILLUSTRATION 4-2

4. **1.5T:** Connect the dummy load and wattmeter (10kW element) to the body output of the cabinet. See Illustration 4-3 or Illustration 4-4 (RF Power Kit).



6. Disable both power monitors by placing the two monitor switches on the front panel of the SSM in the Bypass position.
7. Remove the test window from the Test Switches and Test LEDS on the front of the SSM. See Illustration 4-7.



INSIDE THE TEST WINDOW
ILLUSTRATION 4-7

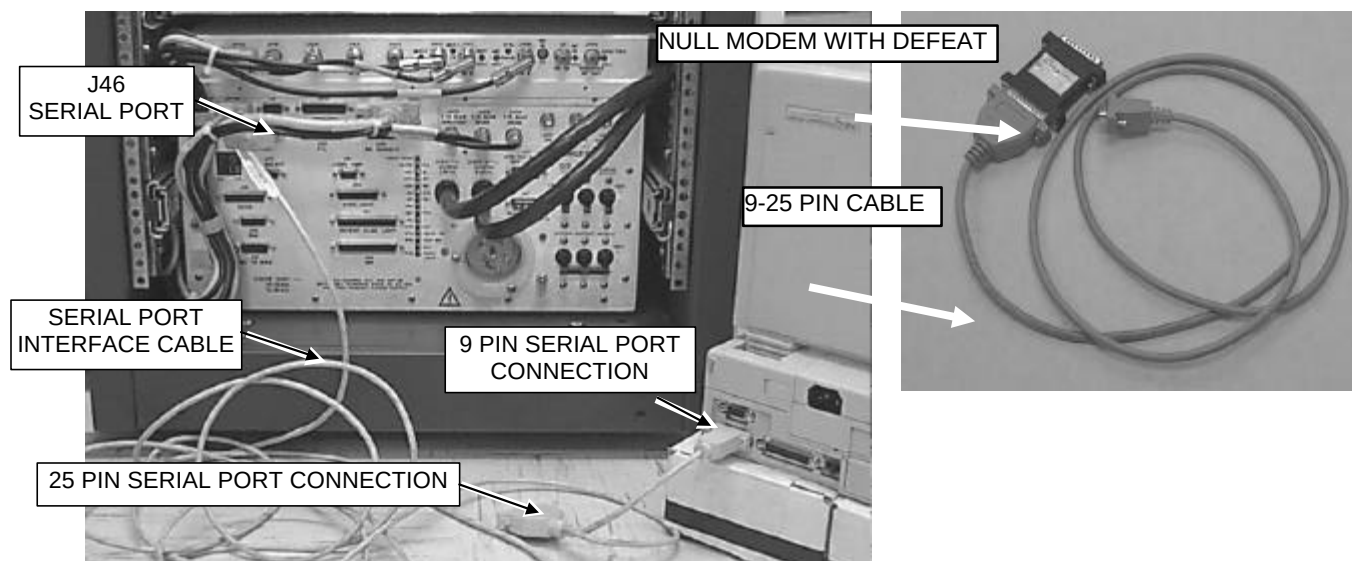
8. Place the first four switches in the order and positions shown in Table 4-1 or for the alternate proprietary procedure, see Table 4-2.

TABLE 4-1
SWITCH POSITIONS

SWITCH NUMBER	POSITION
2	UP
3	DOWN
4	UP
1	UP

TABLE 4-2
ALTERNATE PROPRIETARY PROCEDURE USING LAPTOP

Connect the laptop serial port to J46 on the rear of the SSM, using cable part number 2124497-46 (supplied with every cabinet). See Illustration 4-7. If this cable is not available, use a suitable null modem and 9-25 pin cable adapter. See Illustration 4-7.



LAPTOP CONNECTED TO SSM
ILLUSTRATION 4-7

5- BODY CHECKS

Tests in this section allow the service engineer to selectively force body mode peak power, pulse width, and duty cycle beyond allowable limits.

Note

If using the Alternate Proprietary Procedure Using the Laptop, see Table 5-1. Otherwise move to step 2.

TABLE 5-1
ALTERNATE PROPRIETARY PROCEDURE — LAPTOP SOFTWARE

1. Start the software diagnostic program MONS.EXE on the laptop.

Note

The MONS.exe program if not loaded on the service laptop can be loaded from the MR Service CD-ROM 2187583-2. Insert your latest 2187583-2 CD into the CD ROM drive. Under Windows 95 use the cursor to select "**Start**" and then "**Run**". In the edit box type the following.

d:\mrtools\win95.exe

Follow the instructions of the setup program to install the Service Tools. The MONS.EXE must be run from DOS. In Windows95 perform the following steps:

- 1) **[Start], [Shut Down...], Restart the computer in MS-DOS mode?, [Yes]**
- 2) At the C:\Windows> prompt, type **cd.<ENTER>**
- 3) Type **cd Cclass<ENTER>**
- 4) At the C:\CCLASS> prompt, type **cd erbtec<ENTER>**
- 5) At the C:\CCLASS\ERBTEC> prompt type **mons.exe<ENTER>**

Select **N** for Next Screen, then select **T** for Test Mode Screen. This screen latches a fault to the laptop screen when it occurs, and the amplifier will not shut down. The main screen will display information needed for the tests.

2. At the operator work space, prepare the system for a Body Power Monitor scan using the "Service Protocols" procedure, see Table 5-2.

TABLE 5-2
SCAN PRESCRIPTION - BODY PM CHECKS

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

- A. **[New Pt]**
Id: **geservice** <ENTER>
Name: **pmc**
Weight (Lb.): **300**
Set Patient Protocols to **Service**.
- B. At front enclosure:
Landmark in the Head area—remove any coils.
press **LANDMARK**.
press **MOVE TO SCAN**.
- A. **[New Series]**
At Patient Protocols – select **other**.
- B. In the protocol field, type **o.23.2**<ENTER> (o=Other, 23.2 =series) to load the body protocol
OR select **[o.23] [Series 2] [Accept]**.
[OK] (if required).
- C. **[Save Series]**
- D. **[Prepare to Scan]**.
- E. **[Research Operations]**.
[Setup Params]. Set Plot to **OFF**. Set **TG** to **50**. **[Done]**.
- F. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **dcset** <ENTER>, **255** <ENTER>
CV name: **t3** <ENTER>, **20000** <ENTER>
CV name: **see appropriate Table**<ENTER>, **see appropriate Table** <ENTER>
[Accept].
- F. **[Research Operations] [Download]**.
- G. **[Prepare to Scan]**.
- H. Select **[Manual Prescan]**.

3. Perform body checks as per Table 5-3 (1.5T) or Table 5-4 (1.0T).

Note

All setpoint values are selected by modifying control variables. Instructions are provided in the Scan Conditions column in Table 5-3 (1.5T) or Table 5-4 (1.0T).

Note

Wattmeter reading must be multiplied by cable attenuation factor of cables between the RF amplifier and the wattmeter.

Note

Oscilloscope reading must be multiplied by dummy load and cable(s) attenuation factor between the RF amplifier and the oscilloscope.

5- BODY CHECKS (continued)

TABLE 5-3
BODY CHECKS (1.5T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
BODY PEAK POWER HIGH	[Display CVs] calmode = 2 trig = 7 aset = 120 [Accept] [Download] [Manual Prescan].	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure Power. Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure Power.	Power is within specifications Min: 8793.9W or 69.4dBm Max: 11897.7W or 70.75dBm Nom: 10345.8W or 70.1dBm ☐ Pass ☐ Fail	Set TG to 0 [Done]. Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
BODY PEAK POWER LOW	[Display CVs] aset = 30 [Accept] [Download] [Manual Prescan].	Increase TG until laptop or SSM LED indicates fault occurred on one of the monitors; measure Power. Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure Power.	Power is within specifications Min: 2198.5W or 63.4dBm Max: 2974.5W or 64.7dBm Nom: 2586.5W or 64.1dBm ☐ Pass ☐ Fail	Set TG to 0 [Done]. Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
BODY PULSE WIDTH(PW) HIGH (Min Limit)	[Display CVs]: calmode = 1 p1 = 4750 aset = 255 pwset = 100 [Accept] [Download] [Manual Prescan].	Increase TG (from 0) until Power measures 2 - 3kW. (Note 2)	SENSE LEDs (both blinking), ☐ Pass ☐ Fail	[Done].
BODY PW HIGH (Max Limit)	[Display CVs]: p1 = 5250 [Accept] [Download] [Manual Prescan].	Do not change TG.	Laptop or SSM LED indicates monitor faulted. ☐ Pass ☐ Fail	[Done] Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
BODY PW LOW (Min Limit)	[Display CVs]: p1 = 400 pwset=10 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking) ☐ Pass ☐ Fail	[Done].
BODY PW LOW (Max Limit)	[Display CVs]: p1 = 600 [Accept] [Download] [Manual Prescan].	Do not change TG.	Laptop or SSM LED indicates monitor faulted. ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
BODY DUTY CYCLE (DC) HIGH (Min Limit)	[Display CVs]: calmode = 3 t3=33333 TR_SLOP=0 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), ☐ Pass ☐ Fail	[Done].
BODY DC HIGH (Max Limit)	[Display CVs]: p3 = 4767 [Accept] [Download] [Manual Prescan].	Do not change TG.	Laptop or SSM LED indicates monitor faulted. ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.

5- BODY CHECKS (continued)

TABLE 5-3(continued)
BODY CHECKS (1.5T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
BODY DC LOW (Min Limit)	[Display CVs]: p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), ☐ Pass ☐ Fail	[Done].
BODY DC LOW (Max Limit)	[Display CVs]: p3 = 917 [Accept] [Download] [Manual Prescan].	Do not change TG.	Laptop or SSM LED indicates monitor faulted. ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
Note 1: Wattmeter reading must be multiplied by Cable Attenuation factor from Table 4-1, Set Up and Calibration tab. Note 2: If necessary, increase TG until SENSE LEDs come on. No more than 5 kW should be necessary.				

5- BODY CHECKS (continued)

TABLE 5-4
BODY CHECKS (1.0T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
BODY PEAK POWER HIGH	[Display CVs] calmode = 2 trig = 7 aset = 110 [Accept] [Download] [Manual Prescan].	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure Power. Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure Power.	Power is within specifications: Min: 4100W or 66.1dBm Max: 4500W or 66.5dBm Nom: 4300W or 66.3dBm ☐ Pass ☐ Fail	Set TG to 0 [Done]. Place both monitor switches to RESET , then BYPASS . On the laptop, press 'C' to continue.
BODY PEAK POWER LOW	[Display CVs] aset = 30 [Accept] [Download] [Manual Prescan].	Increase TG until first FAULT LED on; measure Power. Continue increasing TG until other FAULT LED on, measure Power.	Power is within specifications: Min: 1100W or 60.4dBm Max: 1300W or 61.1dBm Nom: 1200W or 60.8dBm ☐ Pass ☐ Fail	Set TG to 0 [Done]. Place both monitor switches to RESET , then BYPASS . On the laptop, press 'C' to continue.
BODY PULSE WIDTH(PW) HIGH (Min Limit)	[Display CVs] calmode = 1 p1 = 3912 aset = 255 pwset = 35 [Accept] [Download] [Manual Prescan].	Increase TG (from 0) until Power measures 2 - 3Kw. (Note 1)	SENSE LEDs (both blinking), BODY LEDs (both ON), FAULT LEDs (both OFF) ☐ Pass ☐ Fail	[Done].
BODY PW HIGH (Max Limit)	[Display CVs] p1 = 4324 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On the laptop, press 'C' to continue.
BODY PW LOW (Min Limit)	[Display CVs] p1 = 1076 pwset=10 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), BODY LEDs (both ON), FAULT LEDs (both OFF) ☐ Pass ☐ Fail	[Done].
BODY PW LOW (Max Limit)	[Display CVs] p1 = 1276 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On the laptop, press 'C' to continue.
BODY DUTY CYCLE (DC) HIGH (Min Limit)	[Display CVs] calmode = 3 t3=33333 TR_SLOP=0 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), BODY LEDs (both ON), FAULT LEDs (both OFF) (Note 2) ☐ Pass ☐ Fail	[Done].
BODY DC HIGH (Max Limit)	[Display CVs] p3 = 4767 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) (Note 2) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On the laptop, press 'C' to continue.

5- BODY CHECKS (continued)

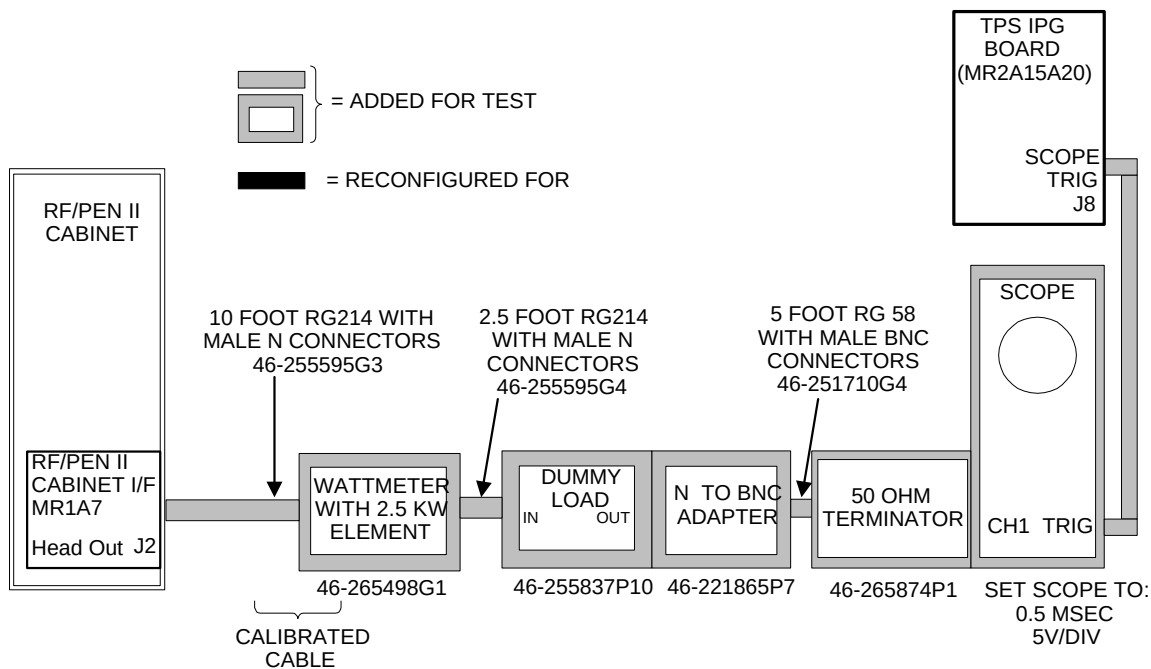
TABLE 5-4 (continued)
BODY CHECKS (1.0T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
BODY DC LOW (Min Limit)	[Display CVs] p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), BODY LEDS (both ON), FAULT LEDs (both OFF) (Note 2) ☐ Pass ☐ Fail	[Done].
BODY DC LOW (Max Limit)	[Display CVs] p3 = 917 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) (Note 2) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On the laptop, press 'C' to continue.
Note 1: If necessary, increase TG until both SENSE and both BODY LEDs come on. No more than 1 kW should be necessary. Note 2: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.				

6- HEAD CHECKS

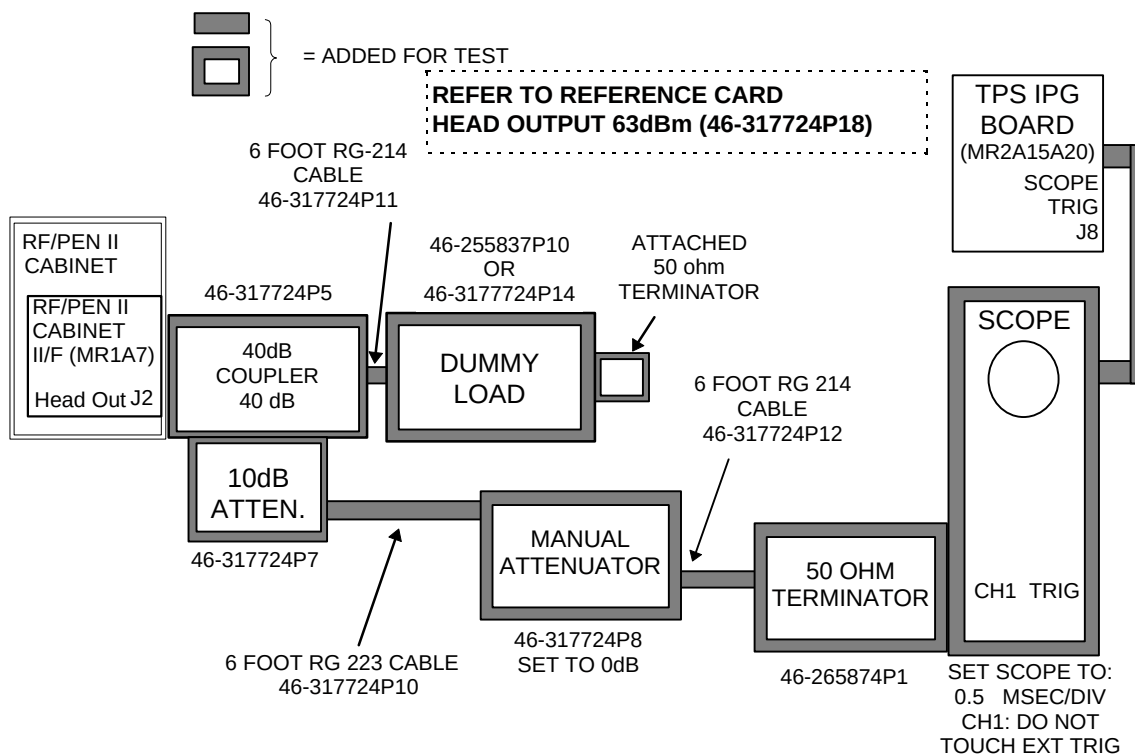
Tests in this section allow the service engineer to selectively force head mode peak power, pulse width, and duty cycle beyond allowable limits.

1. **1.5T:** Connect dummy load and wattmeter (2.5 kW element) to the head output of the cabinet. See Illustration 6-1 or Illustration 6-2 (RF Power Kit).



RECONFIGURATION FOR HEAD CHECKS (1.5T)
ILLUSTRATION 6-1

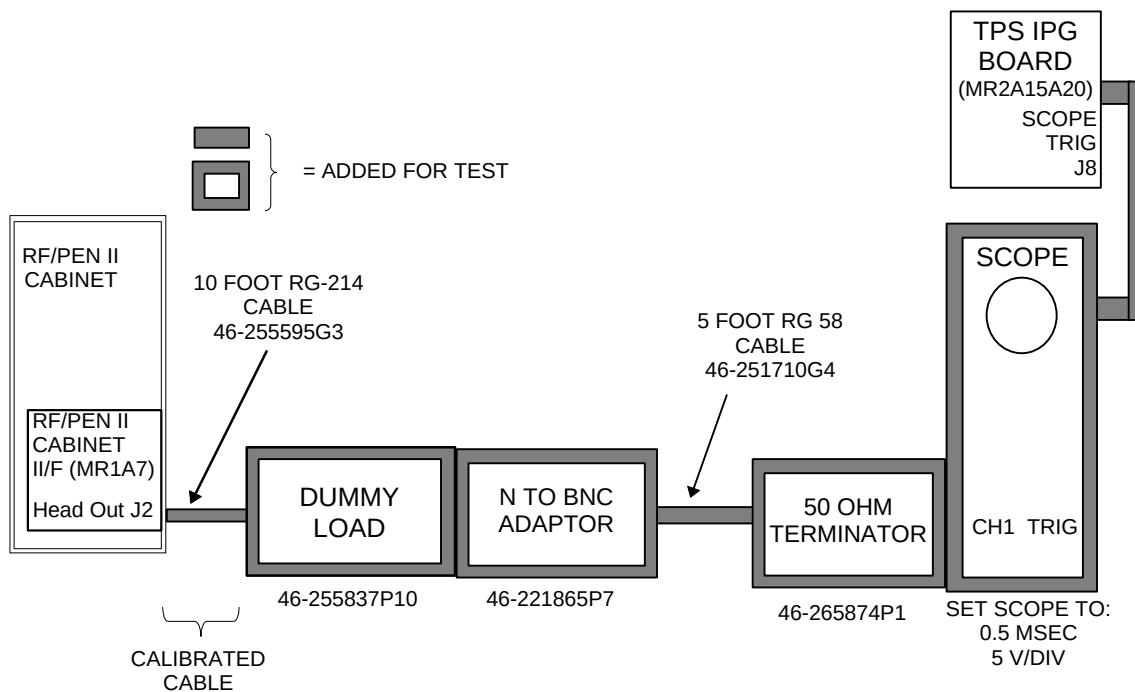
6- HEAD CHECKS (continued)



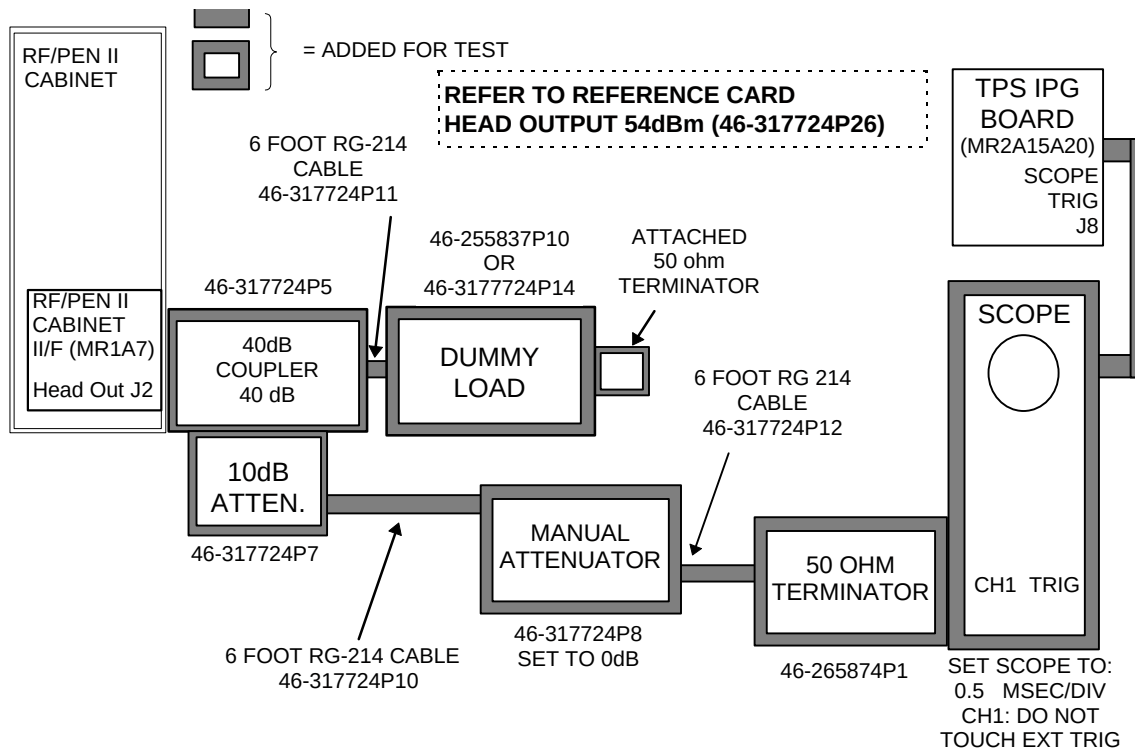
RECONFIGURATION FOR HEAD CHECKS USING RF POWER MEASUREMENT KIT (1.5T)
ILLUSTRATION 6-2

6- HEAD CHECKS (continued)

2. **1.0T:** Connect dummy load to the head output of the cabinet. See Illustration 6-3 or Illustration 6-4.



RECONFIGURATION FOR HEAD CHECKS (1.0T)
ILLUSTRATION 6-3



RECONFIGURATION FOR HEAD CHECKS USING RF POWER MEASUREMENT KIT (1.0T)
ILLUSTRATION 6-4

6- HEAD CHECKS (continued)

3. At the operator work space, prepare the system for a Head Power Monitor scan using the "Service Protocols" procedure, see Table 6-1.

TABLE 6-1
SCAN PRESCRIPTION - HEAD PM CHECKS

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

- A. **[New Pt]**
Id: **geservice** <ENTER>
Name: **pmc**
Weight (Lb.): **300**
Set Patient Protocols to **Service**.
- B. At front enclosure:
Landmark in the Head area—remove any coils.
press **LANDMARK**.
press **MOVE TO SCAN**.
- C. **New Series]**
At Patient Protocols – select **other**.
- A. At Patient Protocols – select **other**.
- B. In the protocol field, type **o.23.1**<ENTER> (o=Other, 23.1 =series) to load the head protocol
OR select **[o.23] [Series 1] [Accept]**.
[OK] (if required).
- C. **[Save Series]**
- D. **[Prepare to Scan]**.
- E. **[Research Operations]**.
[Setup Params]. Set Plot to **OFF**. Set TG to **50**. **[Done]**.
- F. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **see appropriate Table**<ENTER>, **see appropriate Table** <ENTER>
[Accept].
- I. **[Research Operations] [Download]**.
- J. **[Prepare to Scan]**.
- K. Select **[Manual Prescan]**.

4. Perform head checks as per Table 6-2 (1.5T) or Table 6-3 (1.0T).

Note

All setpoint values are selected by modifying control variables. Instructions are provided in the Scan Conditions column in Table 6-2 (1.5T) or Table 6-3 (1.0T).

Wattmeter reading must be multiplied by cable attenuation factor of cables between the RF amplifier and the wattmeter.

Oscilloscope reading must be multiplied by dummy load and cable(s) attenuation factor between the RF amplifier and the oscilloscope.

6- HEAD CHECKS (continued)

TABLE 6-2
HEAD CHECKS (1.5T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
HEAD PEAK POWER HIGH	[Display CVs]: calmode = 2 trig = 7 aset = 120 [Accept] [Download] [Manual Prescan].	Increase Transmit Gain (TG) until, first FAULT LED on, measure Power. Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure power.	Power is within specifications: Min: 1318W or 61.2dBm Max: 1784W or 62.5dBm Nom: 1551W or 61.9dBm ☐ Pass ☐ Fail	Set TG to 0[Done] Place both MONITOR switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD PEAK POWER LOW	[Display CVs]: aset = 30 [Accept] [Download] [Manual Prescan].	Increase Transmit Gain (TG) until, first FAULT LED on; measure Power. Continue increasing TG until other FAULT LED on; measure Power.	Power is within specifications: Min: 330W or 55.2dBm Max: 446W or 56.5dBm Nom: 388W or 55.9dBm ☐ Pass ☐ Fail	Set TG to 0[Done] Place both MONITOR switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD PULSE WIDTH(PW) HIGH (Min Limit)	[Display CVs]: calmode = 1 p1 = 4750 aset = 255 pwset = 100 [Accept] [Download] [Manual Prescan].	Increase TG (from 0) until Power measures 200 - 300W. (Note 2)	SENSE LEDs (both blinking), FAULT LEDs (both OFF) ☐ Pass ☐ Fail	[Done]
HEAD PW HIGH (Max Limit)	[Display CVs]: p1 = 5250 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) ☐ Pass ☐ Fail	[Done] Place both MONITOR switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD PW LOW (Min Limit)	[Display CVs]: p1 = 400 pwset=10 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), HEAD LEDs (both ON), FAULT LEDs (both OFF) ☐ Pass ☐ Fail	[Done]
HEAD PW LOW (Max Limit)	[Display CVs]: p1 = 600 [Accept] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) ☐ Pass ☐ Fail	[Done], Place both MONITOR switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD DUTY CYCLE (DC) HIGH (Min Limit)	[Display CVs]: calmode = 3 TR_SLOP= 0 t3=33333 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), FAULT LEDs (both OFF) (Note 3) ☐ Pass ☐ Fail	[Done]
HEAD DC HIGH (Max Limit)	[Display CVs]: p3 = 4767 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) (Note 3) ☐ Pass ☐ Fail	[Done], Place both MONITOR switches to RESET , then BYPASS . On laptop, press 'C' to continue.

6- HEAD CHECKS (continued)

TABLE 6-2(continued)
HEAD CHECKS (1.5T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
HEAD DC LOW (Min Limit)	[Display CVs]: p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), FAULT LEDs (both OFF) (Note 3) ☐ Pass ☐ Fail	[Done]
HEAD DC LOW (Max Limit)	[Display CVs]: p3 =917 [Accept] [Download] [Manual Prescan]	Do not change TG.	FAULT LEDs (both ON) (Note 3) ☐ Pass ☐ Fail	[Done], Place both MONITOR switches to RESET, then BYPASS. On laptop, press 'C' to continue.
Note 1: Wattmeter reading must be multiplied by Cable Attenuation factor from <i>Dummy Load & Cables Calibration</i> procedure under RF tab. Note 2: If necessary, increase TG until SENSE LEDs come on. No more than 500 Watts should be necessary. Note 3: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.				

6- HEAD CHECKS (continued)

TABLE 6-3
HEAD CHECKS (1.0T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
HEAD PEAK POWER HIGH	[Display CVs]: calmode = 2 trig = 7 aset = 50 [Accept] [Download] [Manual Prescan].	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure power. Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure power.	Power is within specifications: Min: 160W or 52dBm Max: 240W or 53.8dBm Nom: 200W or 53dBm ☐ Pass ☐ Fail	Set TG to 0[Done] Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD PEAK POWER LOW	[Display CVs]: aset = 30 [Accept] [Download] [Manual Prescan].	Increase Transmit Gain (TG) until, first FAULT LED on; measure power. Continue increasing TG until other FAULT LED on; measure power.	Power is within specifications: Min: 100W or 50dBm Max: 136W or 51.3dBm Nom: 117W or 50.7dBm ☐ Pass ☐ Fail	Set TG to 0[Done]. Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD PULSE WIDTH(PW) HIGH (Min Limit)	[Display CVs]: calmode = 1 p1 = 3912 aset = 255 pwset = 35 [Accept] [Download] [Manual Prescan].	Increase TG (from 0) until Power measures 50 - 70W. (Note 1)	SENSE LEDs (both blinking), HEAD LEDs (both ON), FAULT LEDs (both OFF) ☐ Pass ☐ Fail	[Done].
HEAD PW HIGH (Max Limit)	[Display CVs]: p1 = 4324 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD PW LOW (Min Limit)	[Display CVs]: p1 = 1076 pwset=10 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), HEAD LEDs (both ON), FAULT LEDs (both OFF) ☐ Pass ☐ Fail	[Done]
HEAD PW LOW (Max Limit)	[Display CVs]: p1 = 1276 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
HEAD DUTY CYCLE (DC) HIGH (Min Limit)	[Display CVs]: calmode = 3 TR_SLOP=0 t3=33333 p3 = 3900 pwset = 255 dcset = 130 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), HEAD LEDs (both ON), FAULT LEDs (both OFF) (Note 2) ☐ Pass ☐ Fail	[Done].
HEAD DC HIGH (Max Limit)	[Display CVs]: p3 = 4767 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) (Note 2) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.

6- HEAD CHECKS (continued)

TABLE 6-3
HEAD CHECKS (1.0T)

TEST	SCAN CONDITIONS	ADJUSTMENTS (Note 1)	VERIFY	CONCLUSION
HEAD DC LOW (Min Limit)	[Display CVs]: p3 = 750 dcset = 25 [Accept] [Download] [Manual Prescan].	Do not change TG.	SENSE LEDs (both blinking), HEAD LEDS (both ON), FAULT LEDs (both OFF) (Note 2) ☐ Pass ☐ Fail	[Done].
HEAD DC LOW (Max Limit)	[Display CVs]: p3 = 917 [Accept] [Download] [Manual Prescan].	Do not change TG.	FAULT LEDs (both ON) (Note 2) ☐ Pass ☐ Fail	[Done], Place both monitor switches to RESET , then BYPASS . On laptop, press 'C' to continue.
Note 1: If necessary, increase TG until both SENSE and both HEAD LEDs come on. No more than 100 Watts should be necessary. Note 2: Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.				

7- AMPLIFIER SHUTDOWN VERIFICATION

Tests in this section verify that each power monitor is capable of shutting down the amplifier independently of the other and further verify both the logic and relay shutdowns for each channel. Setpoints are not checked in this section.

When one or both of the monitors is in bypass mode, the HV relay shutdown is disabled.

1. Disable power monitor B by placing the monitor B switch on the front panel of the SSM in the Bypass position.
2. At the operator work space, prepare the system for a Amp Shutdown Verification scan using the "Service Protocols" procedure located under **[Tools]** menu on the service methods CD-ROM, or for the alternate proprietary procedure, see Table 7-1.

Note

Verify that the Head Coil has been removed from the patient table before continuing.

TABLE 7-1
ALTERNATE PROPRIETARY PROCEDURE — AMP SHUTDOWN

- | | | | | | | | | |
|---|--------------|------------|-------------|-----------|----------------|----------|-------------|----------|
| <ol style="list-style-type: none">1. Click on [New Pt], and <Enter>.
Id: geservice
Name: amplifier shutdown
verification
Weight (lb): 300
Set Patient Protocols to Service
At front of enclosure, press LANDMARK, then MOVE TO SCAN.2. In the protocol field, type o.23.3 (o=Other, 3=Series) to load the protocol.3. [Save Series]4. Select [Research Operations] with the mouse cursor and the right mouse button. Select [Display CVs]. Highlight the CV Name field. Type in the word dcset. Press <Enter>.
Repeat step 4 to enter the following CVs:
<table><tr><td>dcset</td><td>255</td></tr><tr><td>aset</td><td>30</td></tr><tr><td>calmode</td><td>2</td></tr><tr><td>trig</td><td>1</td></tr></table>
[Accept]. | dcset | 255 | aset | 30 | calmode | 2 | trig | 1 |
| dcset | 255 | | | | | | | |
| aset | 30 | | | | | | | |
| calmode | 2 | | | | | | | |
| trig | 1 | | | | | | | |

With the right mouse button, select **Research Operations**. **Download**. **Prepare to Scan**.

3. **[Manual Prescan]** (Do not exit *Manual Prescan* until step 12.)
4. Increase transmit gain (TG) until wattmeter reads 0.5 to 1kW.

7- AMPLIFIER SHUTDOWN VERIFICATION (continued)

5. Continue increasing TG until monitor A FAULT LED illuminates. Quickly verify the following:

- The FAULT LED on for monitor A (resets automatically after two seconds).
- RF amplifier is shut down and is displaying fault code 84 or 85. (IPER will reset the amplifier 10 to 15 seconds after the power monitor fault clears, i.e., a few seconds after the FAULT LED turns off on the amplifier.)
- Wattmeter reads zero RF power (press FWD PED, then MAX).

Note

Do not perform any operations at the scan touch screen or keyboard while In Place Error Recovery (IPER) is resetting the RF amplifier.

6. While waiting for the RF amplifier to recover, place monitor B in normal mode by placing the monitor B switch on the front panel of the SSM in the Normal position. Now both monitors A and B should be in normal mode.

7. After IPER has reset the RF amplifier, first decrease TG by 50 counts (5dB), then press START SCAN on the operator's console to initiate looping.

8. Increase TG until the FAULT LED for monitor A or B illuminates. Quickly verify the following:

- FAULT LED on for monitor A or B (resets automatically after two seconds).
- RF amplifier is shut down and is displaying fault code 84 or 85. (IPER will reset the amplifier 10 to 15 seconds after the power monitor fault clears, i.e., a few seconds after the FAULT LED goes off on the amplifier.)

Note

Software will log an 84 or 85 fault in the message log.

- Wattmeter reads zero RF power (press FWD PED, then MAX).

Note

Do not perform any operations at the scan touch screen or keyboard while In Place Error Recovery (IPER) is resetting the RF amplifier.

9. While waiting for the amplifier to recover, disable power monitor A by placing the monitor A switch on the front panel of the SSM in the Bypass position.

10. After IPER has reset the RF amplifier, first decrease TG by 50 counts (5dB), then press START SCAN on the operator's console to initiate looping.

7- AMPLIFIER SHUTDOWN VERIFICATION (continued)

11. Increase TG until monitor B FAULT LED illuminates. Quickly verify the following:

- FAULT LED on for monitor B (resets automatically after two seconds).
- RF amplifier is shut down and is displaying fault code 84 or 85. (IPER will reset the amplifier 10 to 15 seconds after the power monitor fault clears, i.e., as soon as the FAULT LED turns off on the amplifier.)
- Wattmeter reads zero RF power (press FWD PED, then MAX).

Note

Do not perform any operations at the scan touch screen or keyboard while In Place Error Recovery (IPER) is resetting the RF amplifier.

12. After IPER has reset the RF amplifier, select **[Done]**.

13. **[Set Up Parameters]** and turn power spectrum on.

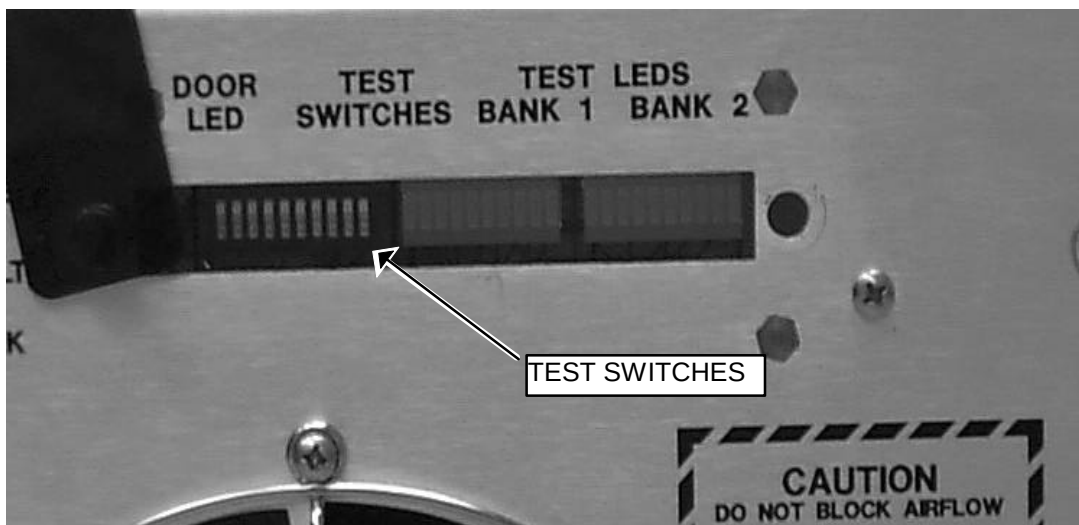
14. In the Rx Manager, select **[End Exam]**.

8- DECLARATION FORM PREPARATION

1. Prepare Direction 2212503, *Signa Horizon Power Monitor Functional Test Declaration*. Refer to *Planned Maintenance Manual* for the Test Declaration form.
2. File completed declaration form per current company policy.

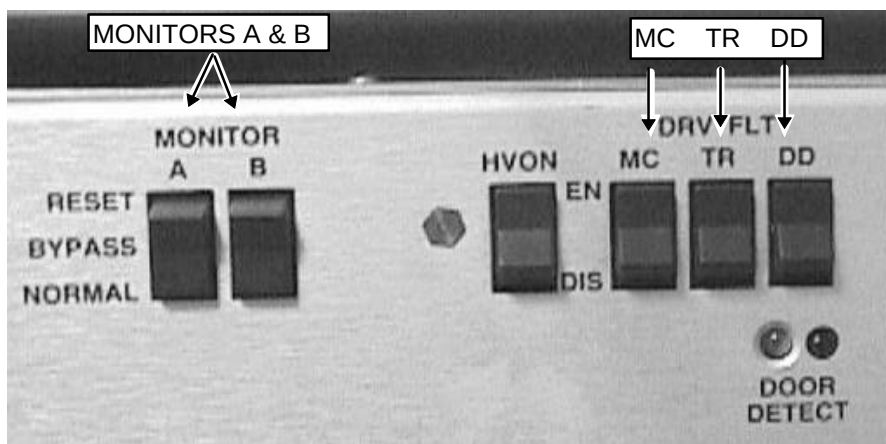
9- SYSTEM RESTORATION

1. In the Test Window place switches 1-4 in the down position and replace the LED cover on the front of the SSM. See Illustration 9-1.



INSIDE THE TEST WINDOW
ILLUSTRATION 9-1

2. On the front panel of the SSM, place the two monitor switches in the Normal position, and place the three DRV FLT switches in the enable faults position (EN). See Illustration 9-2.

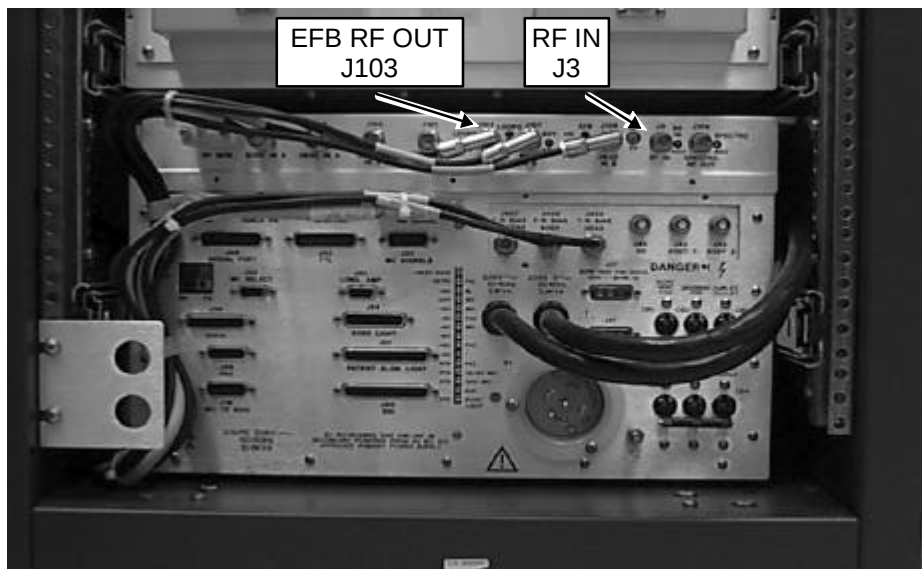


FRONT PANEL SWITCHES ENABLED
ILLUSTRATION 9-2

3. Ensure that the dummy load and wattmeter are disconnected from the rear of the cabinet.
4. Ensure that the head and body heliax cables are reconnected to the rear of the cabinet.

9- SYSTEM RESTORATION (continued)

5. Reconnect BNC cables to RF IN (J3) and EFB RF OUT (J103). See Illustration 9-3.



J103 AND J3 LOCATIONS
ILLUSTRATION 9-3

Note

If using the Laptop perform the following steps to exit MONS.EXE and return to Windows:

- A) Follow screen directions to return to DOS.
 - B) At the DOS C:\CCLASS\ERBTEC> prompt, type **EXIT<ENTER>**.
6. Place the first four DIP switches of the Test Switches in the DOWN position or disconnect laptop computer from J46 on the rear of the SSM.
 7. Complete one head scan satisfactorily.
 8. Complete one body scan satisfactorily.
 9. Replace the front door on the cabinet.
 10. Prepare and file power monitor functional check data sheet according to current company policy. Refer to *Planned Maintenance Manual* for the Test Declaration form.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 10, 1998	Karyl Verdon	Initial Toolbook conversion to Word. (KK)
1	October 19, 1998	K. Keshena	Updated proprietary procedures.
2	Dec. 16, 1998	K. Keshena	Updated Body and Head Tables to reflect SN60932A.
3	Apr. 2, 1998	R. Lambert	Updated head and body protocol tables.
4	Sept. 2, 1999	R. Lambert	Word 97 plus a few minor updates.